MILITARY SPECIFICATION

BOMB, PRACTICE BDU-38/B

This Military Specification is mandatory
for use by the Air Force

1. SCOPE. This specification covers the requirements for one type of practice bomb designated as the BDU-38/B, hereafter referred to as the practice bomb. The practice bomb is designed to simulate retarded and freefall delivery of the parent weapon.

2. APPLICABLE DOCUMENTS. The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-650	Crates, Wood, Open and Covered

Military

MIL-P-116	Preservation, Methods Of
MIL-B-131	Barrier Material, Water-Vaporproof, Flexible
MIL-D-1000	Drawings, Engineering and Associated Equipment
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8591	Airborne Stores and Associated Suspension Equipment, General Design Criteria For
MIL-B-26195	Boxes, Wood-Cleated, Skidded, Load Bearing Base
MIL-W-27000	Weapons, Nuclear, Practice and Training- and Similar Weapons Items, General Specification For

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

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MIL-STD-804	Formats and Coding of Tabulating Aperture Cards for Engineering Data Micro-Reproduction System
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation Of

DRAWINGS

Air Force

66J14360-1	Bomb, Practice BDU-38/B (Retarded)
66J14360-3	Bomb, Practice BDU-38/B (Freefall)

Army

FD24266	Cartridge, Impulse IM-248 AssemblY (Frankford Arsenal)
64B2002	Parachute (Frankford Arsenal)

Commercial

MC1951 Dwg 311932	Spin Rocket (Sandia Corporation)

(Copies of the above documents required by contractors in connection with specific procurement functions should be obtained from the Administrative Contracting Officer or as directed by the Procuring Activity.)

3. REQUIREMENTS

3.1 GENERAL REQUIREMENTS. The requirements of MIL-W-27000 shall apply as requirements. of this specification with the exceptions and additions specified herein. In case of conflict between the two specifications, this specification shall govern.

3.2 PREPRODUCTION. This specification makes provisions for preproduction testing.

3.3 DESIGN AND CONSTRUCTION. The practice bomb and components shall be designed and constructed in conformance to drawing 66J14360 and subsidiary drawings.

3.4 DETAIL DESIGN REQUIREMENTS

3.4.1 Weight. The weight of the practice bomb including parachute shall be 705 $\pm$ 15 pounds.

3.4.2 Parachute Weight. The weight of the parachute including outer tube and deployment plate, shall be 92 $\pm$ 2 pounds.

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3.4.3 Center of Gravity. The Center of Gravity of the assembled bomb, including all government furnished equipment shall be at station 59.6 ± 0.5 inches.

3.4.4 Moment of Inertia. The pitch and yaw moment of inertia about the center of gravity of the assembled bomb shall be 745,500 ± 10% $lb\text{-}in^2$. The roll moment of inertia shall be 15,000 ± 10% $lb\text{-}in^2$.

3.4.5 Location Reference. Station 0 shall be the forward surface of the nose of the practice bomb,

3.4.6 Shear Force. Before relative motion between the inner and outer deployment tubes can begin, the shear pins maintaining the field break at Station 130.663 must shear. These pins shall be sheared by a total shear force of 8,700-10,500 pounds.

3.5 INTERCHANGEABILITY. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not intechangeable, the criteria listed in MIL-D-1000 shall apply and new drawings shall be prepared as necessary.

3.6 REUSABILITY. The unit will be reusable with a minimum of parts replacement. The nature of the deployment operation dictates expenditures of the refurbishment kit.

3.7 ENVIRONMENTAL REQUIREMENTS. The practice bomb shall be capable of meeting the performance requirements of this specification during or after, as applicable, exposure to the following environmental conditions:

a. High Temperature: 418°F for 20 minutes at the nose and leading edges of the fins and 300°F for 20 minutes to all other exterior surfaces of the skin.

b. Low Temperature: -65°F (indefinite storage). -65°F for power source for two hours.

c. Vibration incident to transportation and flight in carrying aircraft.

d. Shock likely to be encountered during transportation or handling.

e. Shock incident to supersonic flight and maneuver of carrying aircraft.

f. Salt spray likely to be encountered in storage or transportation near a large body of salt water.

3.8 ELECTRICAL SYSTEM TEST. The electrical system components shall be capable of satisfactory operation after the environmental tests of this specification are performed.

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3.9 STRENGTH REQUIREMENTS

3.9.1 The assembled bomb shall be capable of withstanding a 15g shock loading applied vertically through the suspension lugs.

3.9.2 The assembled bcmb shall be capable of withstanding a 120,000 pound parachute opening shock loading applied longitudinally through the twelve (12) suspension line support points.

3.9.3 The suspension lugs shall be capable of withstanding a 20g static loading applied vertically on the load bearing surfaces of the lugs.

3.9.4 The lugs shall be capable of passing the load conditions which includes a factor of safety of 50 percent.

3.10 IDENTIFICATION OF PRODUCT. The practice bomb and all parts and assemblies shall be marked in accordance with MIL-STD-130.

3.11 GOVERNMENT FURNISHED PROPERTY. Unless otherwise specified in the contract, the following items shall be furnished to the contractor, upon his request, as Government furnished property for the purpose of fabrication of the practice bomb.

Part Number	Nomenclature	Quantity
64B2002	Parachute	1 per retarded practice bomb
FD24266	Cartridge M248	1 per retarded practice bomb
MC1951	Spin Rocket (Sandia Corp)	1 per freefall practice bomb

3.12 WORKMANSHIP. Workmanship shall be as specified in MIL-W-27000."

4. QUALITY ASSURANCE PROVISIONS

4,1 CLASSIFICATION OF TESTS. Inspection and testing of the practice bomb shall be classified as follows:

a. Preproduction Tests

b. Acceptance Tests

4.2 PREPRODUCTION TESTING

4.2.1 Preproduction Test Samples. The preproduction test units shall be representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity, or when so stated in the contract, at the contractor's facility under the supervision of the procuring activity.

4.2.1.1 Preproduction Test Reports. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion

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of the preproduction tests, prepare a preproduction test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.2.2 Preproduction Tests. Preproduction tests shall consist of all tests described in "4.5 Test Methods."

4.3 Acceptance Tests. Acceptance tests shall consist of the following individual tests.

a. Examination of Product

b. Conformance to Drawings

4.4 Test Conditions. Test conditions shall be as required for each specific test.

4.5 Test Methods. All tests shall be performed with the practice bomb completely assembled.

NOTE: In no case shall live explosive devices be assembled in the bomb being tested. Inert models only shall be used.

4.5.1 Examination of Product. Each practice bomb shall be examined for strict conformance to the requirements of this specification for which tests are not specified. Failure to meet these requirements shall be cause for rejection.

4.5.2 Conformance to Drawings. Each practice bomb shall be inspected for conformance to the drawings, and to the specifications and standards specified therein. Failure to conform to these criteria shall be cause for rejection.

4.5.3 Static and Environmental Testing. The practice bomb shall be subjected to the following static and environmental tests in the listed order:

4.5.3.1 Weight. The practice bomb shall be weighed using a scale calibrated to a standard traceable to the National Bureau of Standards. The weight shall be 705 $\pm$ 15 pounds. Weight not falling within these tolerances shall be cause for rejection.

4.5.3.2 Center of Gravity. The center of gravity of the practice bomb shall be determined by any convenient method. The center of gravity shall be at station 59.6 $\pm$ 0.5 inches. A determined center of gravity not within these tolerances shall be cause for rejection.

4.5.3.3 Moment of Inertia Test. The fixture for this test shall be a simple torsion pendulum or bar with an adjustable hanger which can be attached to the practice bomb. This fixture shall be calibratied by swinging a bar of known weight and calculated moment of inertia. The periods (in seconds/oscillation) obtained from this swinging

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shall, by substitution in the following formula, be used to determine the spring constant of the torsion bar:

$$I_1 = C_z \frac{(P)}{} - I_z$$

Where I_1 = measured inertia (slug ft^2)

C_z = spring constant of torsion bar (ft-lb/radian)

P = period (seconds/oscillation)

I = moment of inertia of hangar

4.5.3.3.1 Test Procedure. The practice bomb shall be attached *to* the hangar and swng, and the formula in the moment of inertia test paragraph shall be used to determine the moments of inertia of the practice bomb. Pitch and yaw moments of inertia about the center of gravity shall be 745,500 ± 10% lb-in^2. The roll moment of inertia shall be 15,000 ± 10% lb-in^2. Determined moments of inertia not within these tolerances shall be cause for rejection.

4.5.3.4 Environmental Tests. The practice bomb shall be subjected to the following environmental tests in the sequence listed:

a. High Temperature Test. Expose unit to 418°F for 20 minutes at the nose and leading edges of the fins and tile entire unit to 300°F for 20 minutes.

b. Low Temperature Test. -65°F (indefinite storage) according to Method 502, Procedure 1, MIL-STD-810.

c. Shock Test. Method 516, Procedure 1, MIL-STD-810.

d. Vibration Test. Method 514, Procedure 1, MIL-STD-810.

4.5.3.5 Electrical System Test. A test circuit as shown in Figure 1, shall be set up and used to perform the electrical system test. The electrical system shall be checked to determine if any of the environmental testing has degraded the electrical system components. The tests shall be performed as follows:

a, Screw lanyard hardware out until switch opens.

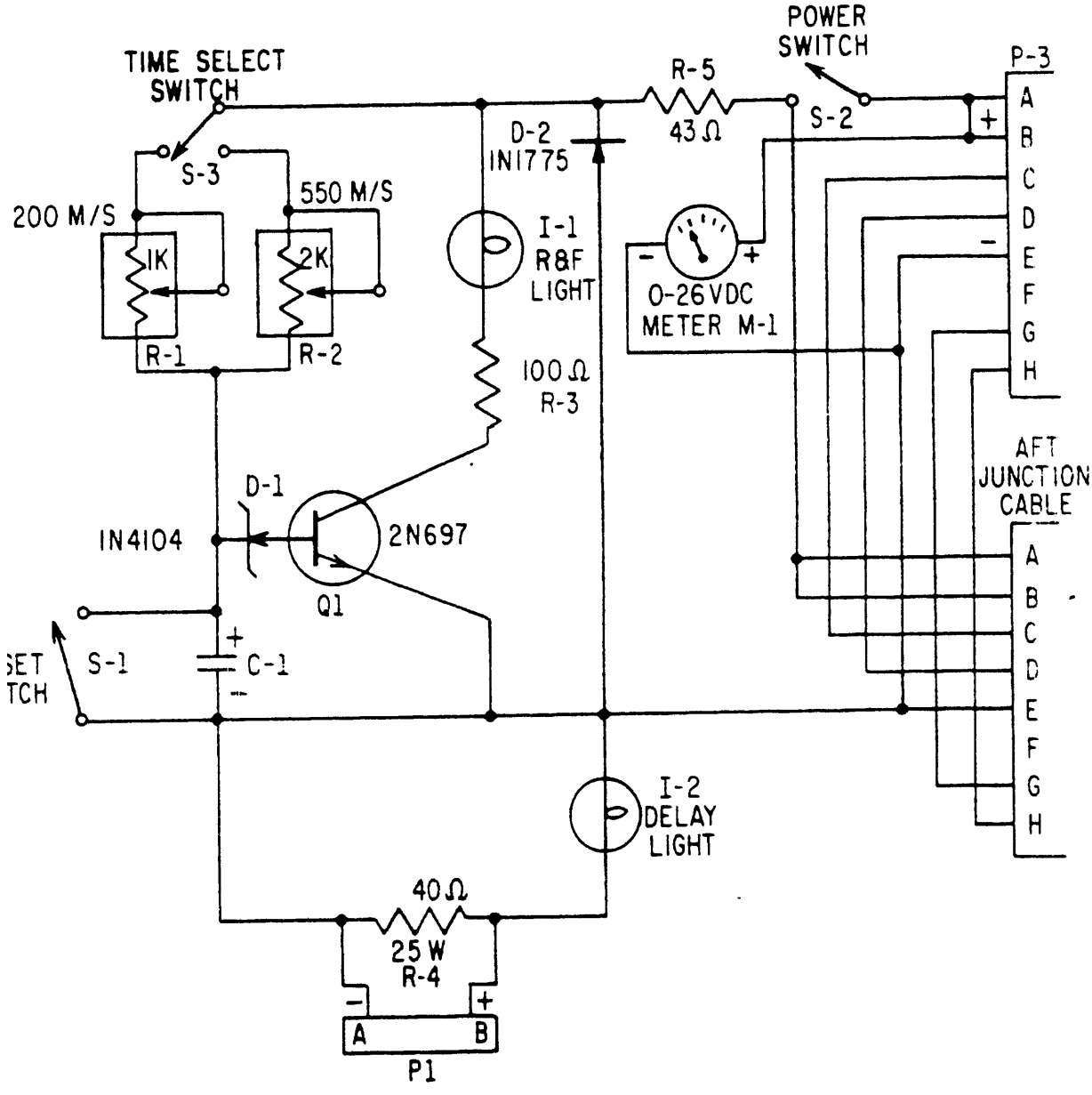
b. Test box power switch OFF.

c. Connect J-3 on cable to P-3 on test box.

d. Observe battery voltage. Voltage should be greater than 16 volts .

e. Connect aft junction cable on box J-3 to tail assembly connector P-3.

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NOTE: After assembly to the box, proper means for inspecting the electrical box circuitry should be used to certify that the 200 ms and 300 ms time delays check out.

FIGURE !=I. ELECTRICAL TEST CIRCUIT (SHEET 1)

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BDU-38/B CHECK OUT BOX PARTS LIST

P-3 PTOOP-12-8P Connector

J-1 PT07P-8-4P Connector

PTOOP-12-8S Aft Junction Connector

JTO2-10-98P Spin Rocket Connector

R-1 1-K Ten Turn Miniature Pot

R-2 2-K Ten Turn Miniature Pot

R-3 100 ohms l/2w Resistor

R-4 40 ohms 25w Resistor

R-5 43 ohms l/2w Resistor

I-1 and I-2 12VDC 20 M2 Lights

M-1 O-25VDC Simpson Wide View Meter

Q-1 2N697 Transistor

S-1 Single Pole Single Throw Spring Loaded to Normally Open

S-2 Single Pole Single Throw

S-3 Double Pole Double Throw Miniature Toggle Switch 8867K4

C-1 200ufd 50VDC

D-1 1N4104 Diodes

D-2 1N1775 Diodes

FIGURE I ELECTRICAL TEST CIRCUIT (SHEET 2)

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f. Connect P-1 on aft cable to J-1 on test box.

g. Set select switch to either the 200 or 550 MS depending on delay already selected.

h. Actuate reset switch.

NOTE: In the next step reference light should come on either at the same instant or before time light. If time light comes on before reference light, the timer should not be used and a new timer should be installed.

i. Turn power on and observe lights. Do not recycle box without waiting 15 seconds.

j. Check lanyard switch by screwing in lanyard hardware. Time light should go out.

k. Turn off power and disconnect test box.

1. Connect J-3 on forward cable to P-3 on aft cable.

Failure to pass the electrical system test shall be cause for rejection.

4.5.3.6 Lug Tests. The lugs shall be tested to the load conditions specified below:

	Load (pounds)		
	Vertical	Axial	Side
Forward 30 inch	23,025	5,115	2,235
	15,375	3,735	10,770
Aft 30 inch	26,205	1,185	2,475
	0	5,115	9,900
Forward 14 inch	39,690	3,900	6,075
	16,275	3,900	15,105

The above quoted values include a safety factor of 150 percent. Failure to pass the lug tests shall be cause for rejection.

4.5.3.7 Shear Force. Tail Cap Static Pull Tests. The after tail cap shall be pulled statically to check the shear force required to separate it from the unit at station 130.663. The shear pins shall be sheared by a total shear force of 8,700-10,500 pounds. Failure to pass the tail cap static pull tests shall be cause for rejection.

4.5.3.8 Fin Tests. The fins and tail section shall be tested to withstand the following load: A 2200 pound load applied to each of two fins at Station 123.11 and 3.35 inches from the bomb skin. The loads

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shall be applied simultaneously, one clockwise and one counterclockwise with the bomb firmly fixed at the center section.

4.5.3.9 Shroud Ring Static Pull. The assembled bomb shall be tested to withstand a load of 120,000 pound parachute opening shockloading applied longitudinally through the twelve (12) suspension line support points. The load should be applied in 10 percent increments until failure occurs or the maximum load of 120,000 pounds is obtained.

5. PREPARATION FOR DELIVERY

5,1 Levels of Preservation and Packaging

5.1.1 Levels A, B, and C. Bomb, Practice BDU-38/B shall be preserved in accordance with MIL-P-116, Method III.

5.2 Levels of Packaging

5.2.1 Level A. Bomb, Practice BDU-38/B, preserved to meet 5.1.1 shall be packed in exterior type containers meeting MIL-B-26195, Type I, Style A, B, or C; Class 1 or 2. As far as practicable, exterior Containers shall be of uniform shape and size be of minimum cube and tare consistent with the protection required. Containers shall be closed and strapped in accordance with the applicable container, specification or appendix thereto.

5.2.2 Levels B and C. Bomb, Practice BDU-38/B, preserved and packed *to* meet 5.1.1 shall be in domestic type exterior containers to meet PPP-C-650. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practicable, exterior containers shall be of uniform shape and size, with the protection required. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.3. Physical Protection. Cushioning, blocking, bracing, bolting, and testing to determine adequacy as required shall be in accordance with MIL-P-7936.

5.4 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be

BOMB, PRACTICE BDU-38/B

6. NOTES

6.1 Intended Use. the BDU-38/B Practice Bomb is intended for use in training aircrews in delivery of the free fall and retarded version of the parent weapon. The practice bomb with the use of spare parts is reusable.

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6.2 Ordering Data. Procurement documents. should specify the following:

a. Title, number and date of this specification.

b. Number of spare parts kits to be furnished if required.

NOTE: These items may be ordered separately.

c. Provisions should be made for making the Parachute Deployment Assembly (PN 66D14398) available to the Government. The Government parachute packing activity will then send the parachute assembly either back to the manufacturer or on to the using agency.

d. Whether preproduction tests will be conducted by the procuring activity.

e. Conditions under which Government loaned property will be made available tc the contractor.

f. Levels of preservation, packaging and packing.

6.2.1 Cartridge Kits. The M248 Cartridges should be procured separately.

6.2.2 Preproduction Test Samples. It is expected that the contract or purchase order will specify that a minimum of two units representative of the production practice bomb will be submitted to the procuring activity for test and evaluation. The invitation for bids, contract or purchase order should specify the point of inspection for these tests.

6.3 Production Delivery. No production practice bomb should be delivered prior to full approval of the preproduction samples.

6.4 Definitions

a. Preproduction Tests. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. Acceptance Tests. Acceptance tests are those tests accomplished on representative smamples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

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